

Polyaniline/Li-Cu Substituted Cobalt Ferrite Composites as Electromagnetic Absorbers in L- and S- Microwave Bands

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Abstract—In this paper, the electromagnetic characterization of absorbing materials was performed using a coaxial transmission line technique. Polyaniline/Li-Cu substituted cobalt ferrite composites were evaluated as microwave absorbers through measurements conducted with a vector network analyzer. The study focused on determining key performance parameters, including absorbance, transmittance, reflection loss, and shielding effectiveness.

Index Terms—Absorbers, Ferrite, Shielding effectiveness, Absorbance, Transmittance.

I. INTRODUCTION

Over the past two decades, there has been increasing interest in the development of radiation-absorbing materials. Among these, spinel-type ferrites offer key advantages, including ease of synthesis, mechanical stability, and chemical inertness. Their magnetic properties can be tailored by adjusting their chemical composition, enabling the tuning of microwave interaction frequencies —as demonstrated, for example, by doping Ni-Zn ferrite with yttrium [1], or by combining spinel and hexagonal ferrites in composite systems. Recent studies have also explored composites based on polyaniline (PANI) and spinel ferrites as electromagnetic absorbers [2]. The synergistic effect between PANI's dielectric loss and the magnetic loss of ferrites enhances impedance matching [3], resulting in broadband absorption, particularly across the 0.5–4.0 GHz range (L and S bands), which is highly relevant for telecommunication applications. Electromagnetic absorber (EMA) characterization typically involves either free-space or guided-wave measurement techniques [4], both relying on a vector network analyzer (VNA) operating within the target frequency band. In this study, a coaxial transmission line setup was used to measure the scattering parameters (*S*-parameters) of the samples under test [4].

The incident power of electromagnetic (EM) radiation can be divided into three parts when EM waves strike the absorbing materials: reflection, absorption, and transmission. The interaction of the EM wave with the molecular and electronic structure of the material will produce heat in the sample that can transform incident EM waves into thermal energy and result in energy absorption [5]. This absorption depends on magnetic and dielectric losses [5]. The theoretical expressions used in this paper are defined in Ref. [2]. The reflectance is $R = |S_{11}|^2$; the transmittance is $T = |S_{21}|^2$ and the absorbance is $A = 1 - R - T$.

The total shielding effectiveness (*SE*) is defined by the relation between the incident E_i and the transmitted E_t electric field intensities (in dB) in the device under test (DUT) inside the transmission line [6]:

$$SE(dB) = -10\log_{10}(T) = 20\log_{10}\left(\frac{E_i}{E_t}\right)$$

II. MATERIALS AND METHODS

Li-Cu(II) substituted cobalt ferrite microparticles of compositions $\text{Co}_{1-x}\text{Li}_x\text{Fe}_{2-x}\text{Cu}_x\text{O}_4$ ($x = 0, 0.01, 0.02, 0.05, 0.1$) were synthesized by calcination of their precursors (6 h at 1100°C). Substitution with Li and Cu(II) ions caused a significant increase in ferrite saturation magnetization, from 77.0 emu/g in CoFe_2O_4 ($x = 0$) to 82.5 emu/g for $x = 0.02$. Polyaniline (PANI) was prepared by oxidation of aniline with FeCl_3 in a hydrochloric solution at room temperature [7]. PANI/ferrite composites (labeled PCF0 to PCF4 for $x = 0$ to $x = 0.1$, respectively) were obtained by synthesis of PANI within suspensions of the ferrite powders. The PANI:ferrite mass ratio in the composite powders was 20:1 [7]. PANI/ferrite composites were compacted into pellets and measured in a teflon sample holder (external diameter = 12.5 mm; internal

diameter = 5.5 mm; thickness = 2.5 mm) inserted in the transmission line.

III. EXPERIMENTAL RESULTS

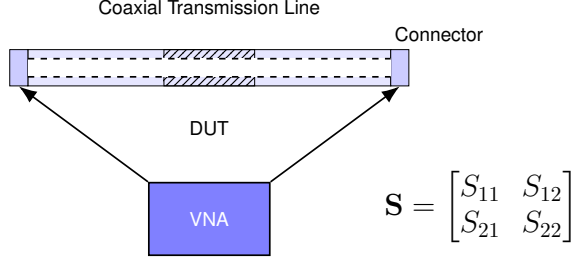


Fig. 1. S-parameter measurement setup using a coaxial transmission line and a vector network analyzer (VNA).

The dispersion parameters or S parameters were measured with the DUT within a coaxial transmission line, as shown in Figure 1. The measurements of Reflectance, Absorbance and Total Shielding versus frequency are shown in Figures 2, 3 and 4, respectively.

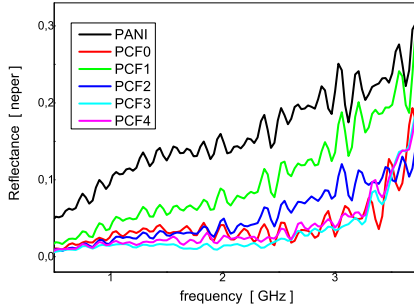


Fig. 2. Reflectance vs frequency.

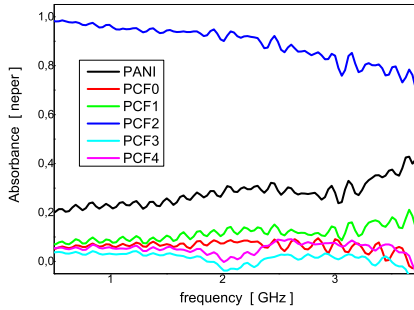


Fig. 3. Absorbance vs frequency.

IV. CONCLUSIONS

All the composite powders, including the pure PANI sample, exhibited low reflectance values —below 0.2 Np— across the

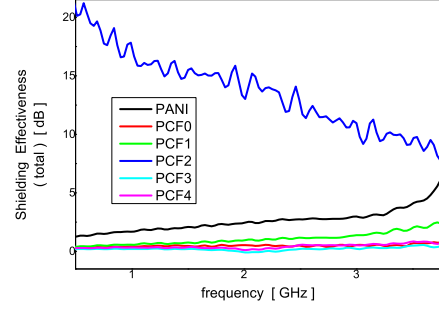


Fig. 4. Total Shielding effectiveness vs frequency.

entire frequency range. Similar results have been reported in previous studies on PANI/ferrite composites [3]. This indicates that the primary contribution to the overall shielding effectiveness (SE) is likely due to electromagnetic wave absorption rather than reflection. Notably, the absorbance was particularly significant for the $x = 0.02$ Li-Cu substituted ferrite composite (sample PCF2), exceeding the absorbance observed in the pure PANI sample. This enhanced absorption leads to substantially higher shielding effectiveness values for PCF2 —ranging from 6 to 18 dB greater than those of PANI— across both the L and S frequency bands. These results suggest that the ferrite of composition $\text{Co}_{0.98}\text{Li}_{0.02}\text{Fe}_{1.98}\text{Cu}_{0.02}\text{O}_4$ is a highly promising candidate for effective electromagnetic absorption within this frequency range.

ACKNOWLEDGMENTS

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